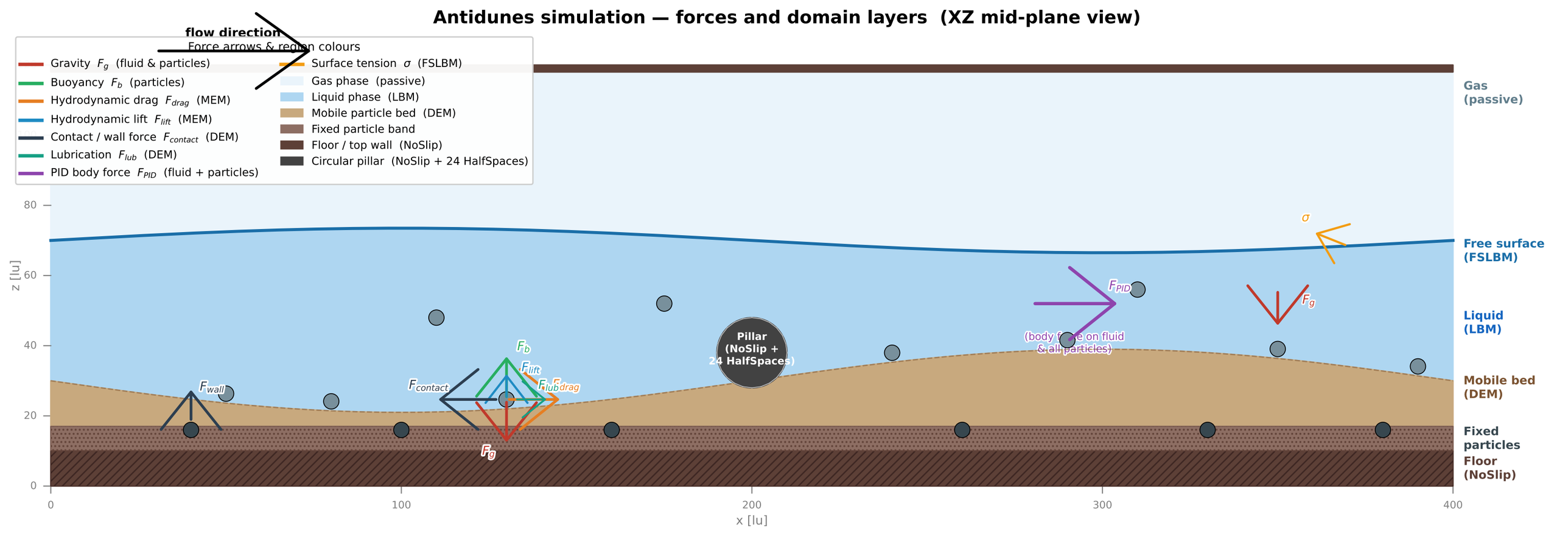

Antidunes simulation — forces and domain layers (XZ mid-plane view)
flow direction
Force arrows & region colours
Gravity F_g (fluid & particles)
Buoyancy F_b (particles)
Hydrodynamic drag F_{drag} (MEM)
Hydrodynamic lift F_{lift} (MEM)
Contact / wall force $F_{contact}$ (DEM)
Lubrication F_{lub} (DEM)
PID body force F_{PID} (fluid + particles)
Surface tension σ (FSLBM)
Gas phase (passive)
Liquid phase (LBM)
Mobile particle bed (DEM)
Fixed particle band
Floor / top wall (NoSlip)
Circular pillar (NoSlip + 24 HalfSpaces)
Gas (passive)
Free surface (FSLBM)
Liquid (LBM)
Mobile bed (DEM)
Fixed particles
Floor (NoSlip)
Pillar (NoSlip + 24 HalfSpaces)
F_{wall}
$F_{contact}$
F_b
F_{lift}
F_{lub}
F_{drag}
F_g
F_{PID}
(body force on fluid & all particles)
σ
z [lu]
x [lu]
0
20
40
60
80
0
100
200
300
400